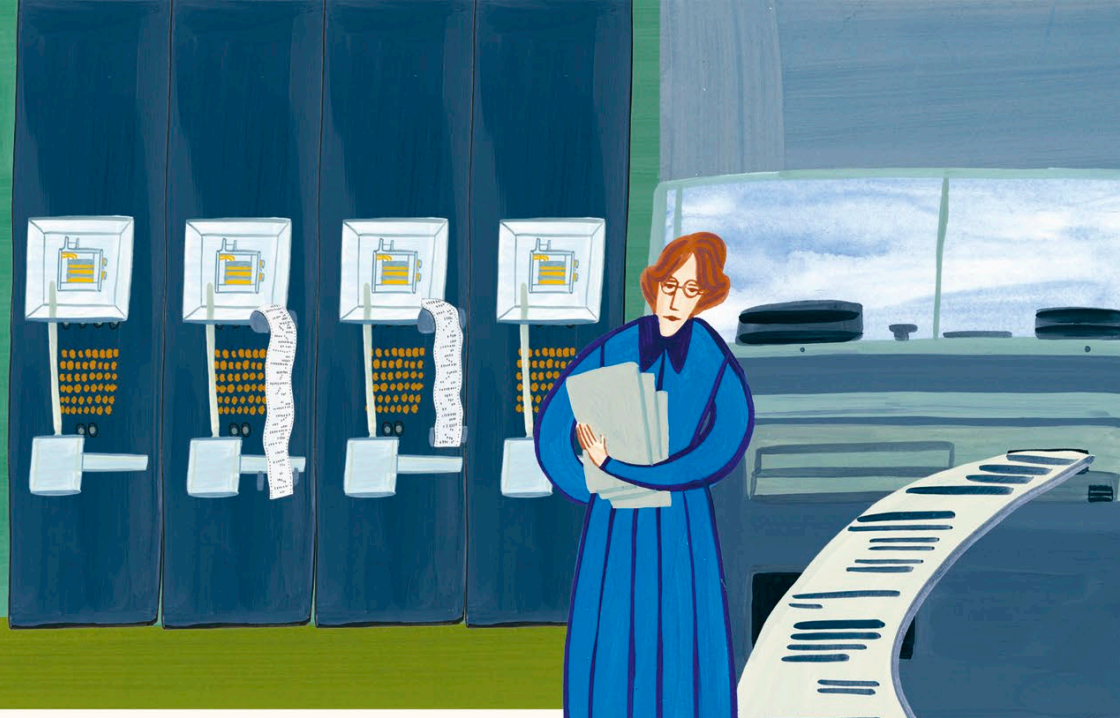


Harvard Mark I

At Harvard University, Grace began working with with one of the world's very first computers—the Harvard Mark I. It was so huge that it filled an entire room, and it performed complicated calculations that were needed to design top-secret weapons during the war. Grace learned to program the Mark I by translating numbers and mathematical symbols into binary code.



Brilliant binary

After the war, Grace helped to develop new and better computers, including the first-ever completely electronic computer. She led a team to develop the first computer language compiler, which could quickly perform the boring job of translating mathematical code into binary.

Next, Grace and her team developed a programming language that used words rather than just numbers and symbols. Again, a compiler did the job of translating the words into binary instructions that the machine could understand. This made it far easier for people who weren't mathematicians to work with computers. The age of computers had begun.

Soon, many different programming languages were being developed. Grace focused on a language called COBOL. She helped to turn it into the most widely used computer language in the world.

The missile destroyer USS Hopper was named after Grace. She was the oldest serving officer in the US Navy when she retired at 79 years old.



Grace loved trying new things. Here, she is coding instructions on punch tape.

